



Lehrstuhl für Data Science

Beyond Fine-Tuning: An Evaluation of Agentic and Actor-Critic LLM Pipelines for Hierarchical Narrative Detection

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dataset also includes a generic Other category for documents that do not fit any predefined narrative, which accounts for a substantial portion of the data and requires models to learn a negative class boundary against all defined narratives. The most prevalent narratives are dominated by war-related topics (e.g., ‘URW: Discrediting Ukraine’, ‘URW: Discrediting the West, Diplomacy’), highlighting the dataset’s focus on high-stakes domains. This sparsity poses a significant challenge for supervised models, which can become biased towards majority classes and fail to generalize to rare but meaningful narratives.

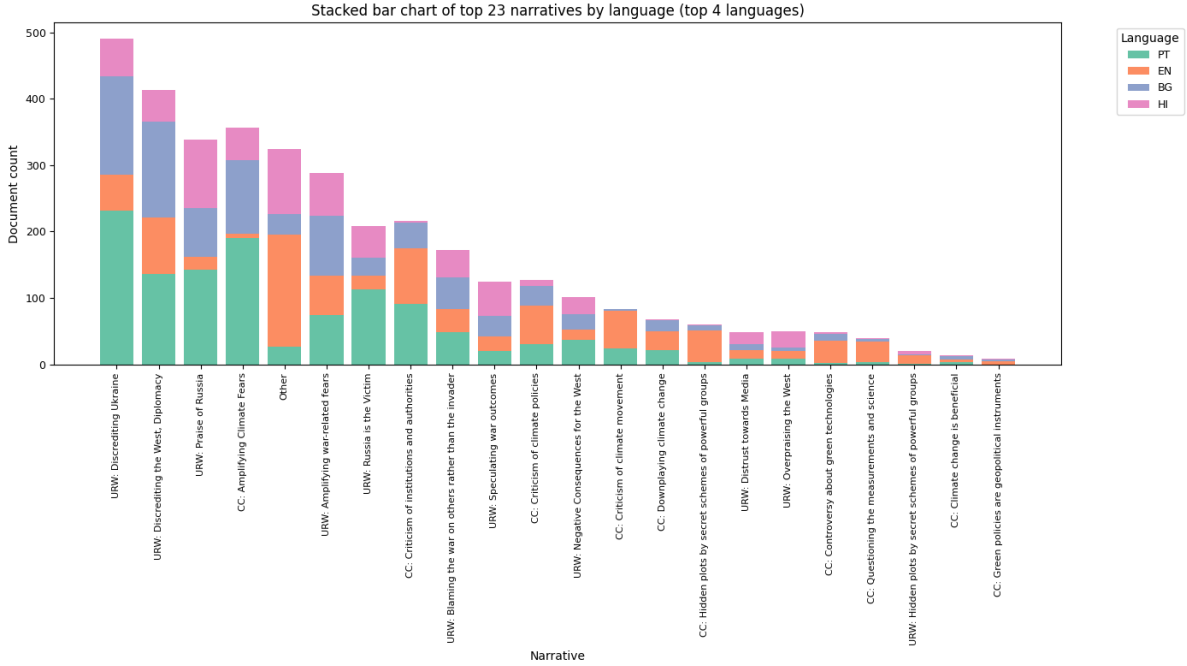


Figure 3.1: Long-tailed distribution of narrative labels in the SemEval-2025 training set. The bar chart shows the frequency of each coarse narrative, illustrating that a small number of narratives account for a large fraction of annotations while many narratives have very few examples. This severe imbalance motivates the use of zero-shot and evidence-grounded LLM approaches that do not rely on large per-class training counts.

Structured Label Co-occurrence. Despite this imbalance, the relationships between labels are not random but exhibit meaningful patterns. Co-occurrence analysis reveals that while most pairs of narratives rarely appear together, certain thematically-linked pairs exhibit strong associations. For example, documents often combine *attack* and *counter-attack* narratives (e.g., discrediting Ukraine together with discrediting the West), or pair attacks on an adversary with *praise of allies* (e.g., discrediting Ukraine alongside praise of Russia). Other recurrent patterns reflect causal framing, such as linking sanctions to economic hardship, or pairings between diplomatic failure and critiques of legitimacy. Military operations are frequently co-annotated

with victim/threat framings, and conspiracy claims commonly appear together with explicit attribution to external actors. These recurring narrative couplings illustrate that co-occurrence captures structured discourse strategies in which labels reinforce one another. More advanced analysis further reveals directional dependencies (i.e., $P(B | A) \neq P(A | B)$).

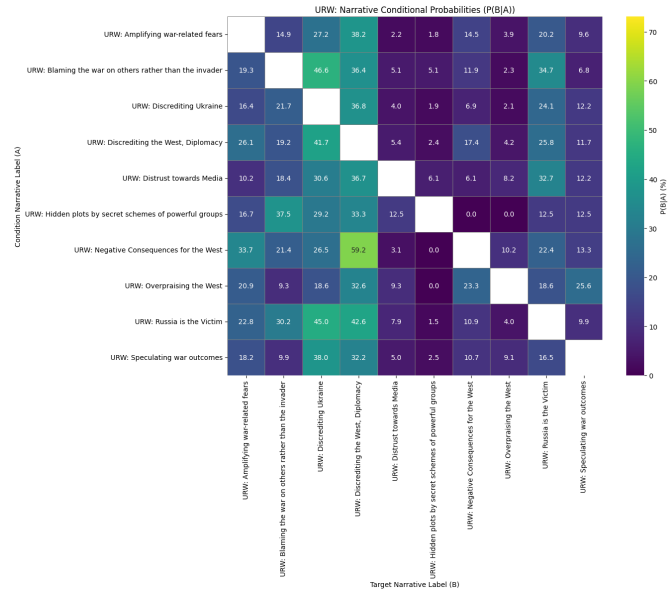
Language-Specific Patterns. These structural complexities are further nuanced by cross-lingual variations. The dataset exhibits notable variation across languages, with the average number of labels per document (label density) highest in Portuguese (≈ 3.04) and lower in Hindi (≈ 1.79), suggesting potential differences in source document complexity or annotation practices. Furthermore, the prevalence of specific narratives and their co-occurrence patterns differ across languages. This cross-lingual heterogeneity underscores the importance of using language-aware methods or, at a minimum, explicitly validating a model’s performance on each language.

Document Length Heterogeneity. Finally, the dataset presents additional challenges through its variable document lengths. The corpus exhibits high variability, with a mean of approximately 404 words, a median of 371 words, and a range spanning from 45 to 4,422 words. This heterogeneity necessitates the use of effective truncation or long-text handling strategies in model design to manage computational constraints without losing critical context.

3 Task and Data Analysis



(a) CC narratives heatmap



(b) URW narratives conditional graph

Figure 3.2: Co-occurrence / conditional-probability visualizations for the most frequent narrative groups. Top: heatmap for CC narratives (normalized cell values). Bottom: conditional-probability visualization for URW narratives (directional). These paired views help identify both symmetric and directional dependencies among narratives.

4 System architectures and methods

To address the challenges of hierarchical multi-label classification presented by the SemEval-2025 dataset, we designed and evaluated three distinct architectural paradigms. We began by establishing a supervised baseline using a fine-tuned Transformer model. We then developed two zero-shot systems that leverage the advanced reasoning capabilities of Large Language Models (LLMs): a parallel agentic framework and a sequential, traceable actor-critic pipeline. This section details the design of each architecture.

4.1 Architecture 0: Transformer Baseline

To establish a robust supervised baseline, we developed a custom Transformer-based architecture explicitly designed to model the two-level taxonomy of the task. Instead of treating the label space as flat, our Multi-Head mDeBERTa for Hierarchical Classification directly encodes the parent-child relationships into its structure. This approach represents a state-of-the-art technique for Hierarchical Multi-Label Classification (HMLC), moving beyond soft regularization penalties to a more principled, hard-coded hierarchical design.

4.1.1 Core architecture and model choice

The foundation of our model is **mDeBERTaV2** (microsoft/mdeberta-v2-base) [He+21], a powerful Transformer encoder chosen for its architectural innovations that are particularly relevant to nuanced understanding tasks. While mDeBERTaV2 is a strong general-purpose model, its **disentangled attention mechanism** provides a compelling theoretical advantage for our specific problem.

Standard Transformer attention mechanisms [Vas+17] compute attention weights based on a single vector that combines both a token’s content (what it is) and its relative position (where it

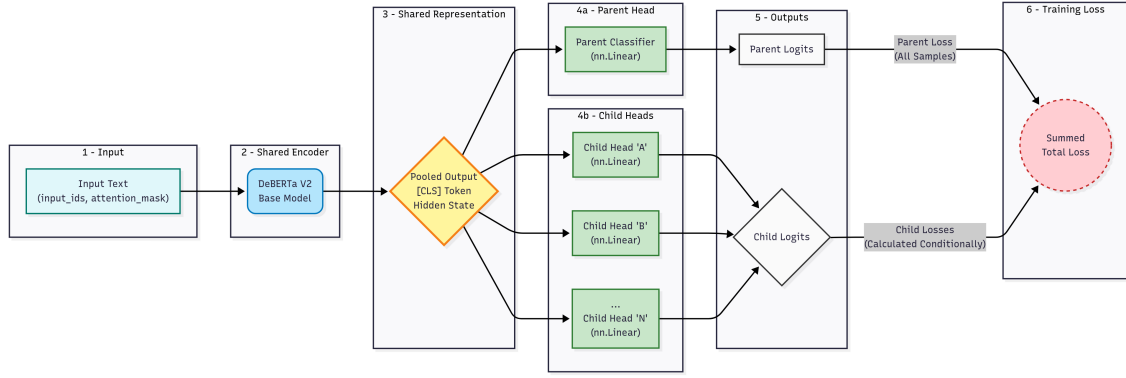


Figure 4.1: Multi-Head mDeBERTa architecture for hierarchical narrative classification. The shared encoder produces a pooled representation that is consumed by a parent-level classifier and multiple conditionally-activated child heads; training aggregates parent loss across all samples while child losses are computed conditionally.

4.1.5 Addressing data scarcity and class imbalance

Data pre-processing

Overview. Now, a critical first step in our methodology was a simple data pre-processing pipeline. As the data is already scarce, we cannot afford to have mDeBERTa learn from noisy inputs. The raw documents, sourced from online news articles, contained non-content artifacts typical of web scrapes, such as residual HTML tags, navigation links, and boilerplate phrases (e.g., “Share this article”). This noise both dilutes the core narrative content and consumes a valuable portion of the model’s input context window (mDeBERTa uses up to 512 tokens).

Cleaning operations. The pipeline aimed to isolate the main body of each article and produce clean, normalized text. Typical operations included removing HTML and scripting remnants, filtering common boilerplate phrases, and normalizing whitespace to ensure consistent formatting. These steps raise the signal-to-noise ratio of the inputs and help models focus their representational capacity on semantically relevant content.

Implementation detail. Rather than constructing a suite of regular expressions or deterministic heuristics, we found **Qwen3-4b-Thinking-2507** to be a highly effective and low-effort option for this task. The LLM was used to identify and delete non-content artifacts while preserving the surrounding text verbatim; in practice, this approach removed navigation elements and calls-to-action embedded inside paragraphs while keeping the original article intact.

Multilingual data augmentation via translation

Motivation. The primary challenge in training the fine-tuning baseline is the severe scarcity of training examples, particularly for the tail-end classes. While the dataset is multilingual, the number of documents in any single language is modest. To significantly expand the training corpus and expose the model to more diverse linguistic expressions of the same narratives, we implemented a data augmentation strategy centered on high-fidelity translation.

Methodology. We leveraged the state-of-the-art translation capabilities of a large language model (Gemini 2.5 Flash) to augment our training data. Our process was as follows:

1. **Translation to a high-resource language:** For each document in the non-English training sets (Bulgarian, Portuguese, Hindi, and Russian), we generated a corresponding English version. We chose English as the target language due to the superior performance of modern LLMs in translating into English, which helps preserve nuanced meaning, tone, and authorial intent.
2. **Corpus expansion:** The newly translated English documents were then added to the training set alongside the original, untranslated documents. This procedure effectively doubles the size of our non-English training data; for example, the 399 Portuguese documents yielded an additional 399 English translations, both used for training.

Intended outcome. This augmentation strategy provides two key benefits. First, it dramatically increases the volume of training data, which helps mitigate overfitting for a large model like mDeBERTa. Second, by training on both the original and translated versions, the multilingual mDeBERTa model is encouraged to learn language-invariant representations of the narratives: it learns to associate a specific narrative with both its original linguistic expression (e.g., in Portuguese) and its English equivalent, pushing it to focus on the core semantic concepts rather than surface-level lexical features. The resulting classifier should be more robust and generalizable across languages.

Handling the “Other” Class

Motivation. The “Other” class represents a substantial portion of the dataset ($\approx 8\%$) and presents a unique modeling challenge. Unlike defined narrative labels, it signifies the *absence* of any specific narrative and, in the training data, never co-occurs with other labels. Including

- **Narrative and Sub-narrative Agents (Specialists).** For every label in the two-level taxonomy we instantiate a dedicated specialist agent. Each such agent is initialized with a system prompt containing the label's official definition and examples, and its single task is to decide whether its assigned narrative is present in the document.

This multi-agent design enables strong specialization: each agent focuses its reasoning on a narrow, well-defined sub-problem and collectively contributes to the final, complex classification.

The *propose and refine* classification workflow

The classification of a document proceeds through a structured, multi-stage workflow that mimics a "propose-and-refine" strategy, as illustrated in Figure 4.2. Instead of a simple broadcast, the system uses Manager Agents at each level of the hierarchy to act as initial, coarse-grained classifiers, whose proposals are then validated by specialized agents.

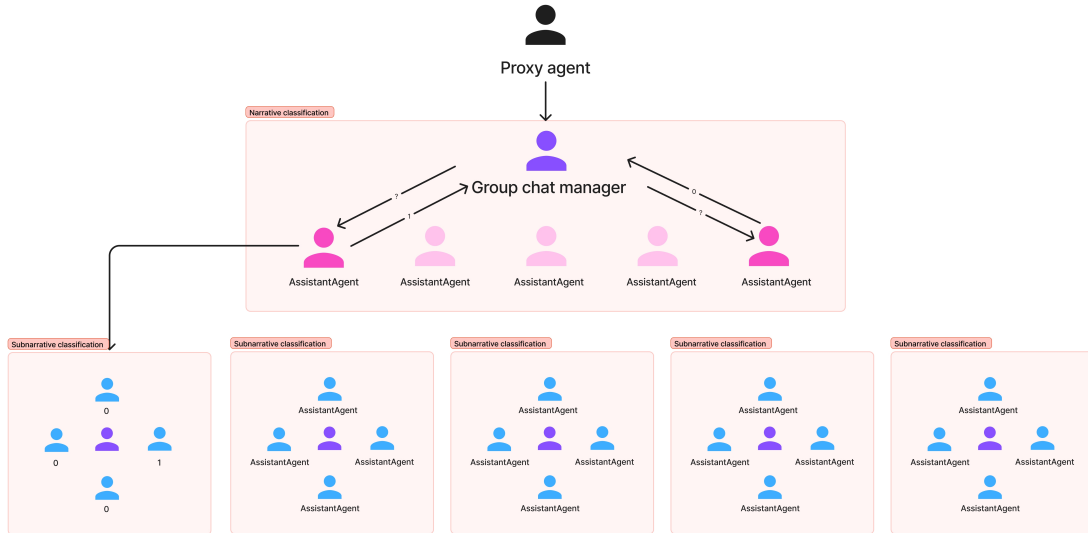


Figure 4.2: AutoGen multi-agent architecture showing the hierarchical propose-and-refine workflow. The Proxy Agent initiates the process, the Group Chat Manager coordinates narrative-level classification, and separate sub-managers handle sub-narrative classification for each confirmed parent narrative.

Stage 1: Narrative-Level Proposal and Refinement Upon receiving a document from the User Proxy, the top-level Manager Agent initiates the classification. It first performs an

internal, high-level analysis of the text to identify a small subset of the most plausible parent narratives. Functioning as a “loose” multi-label classifier, the Manager’s goal is to propose candidate narratives rather than make a final decision.

Once this initial set of candidates is identified, the Manager Agent initiates a “group chat” and invites only the Narrative Agents corresponding to its proposed narratives. It then dispatches the document to this selected group, and each specialist agent is tasked with refining the Manager’s proposal. The agents perform a more detailed, focused analysis and return a definitive binary decision (1 to confirm the proposal, 0 to reject it).

Stage 2: Sub-narrative-Level Proposal and Refinement The process is then repeated for the sub-narrative level. For each narrative confirmed in Stage 1, a dedicated, second-level Manager Agent responsible for that narrative’s children is activated. This sub-manager, acting as another “loose” classifier, analyzes the document to propose the most likely sub-narratives within its specific domain.

Following the same pattern, it creates a new group chat with only the Sub-narrative Agents corresponding to its proposals. These agents then perform the final refinement step, either confirming (1) or rejecting (0) the sub-manager’s hypothesis. This hierarchical and nested manager-specialist structure allows the system to progressively narrow its focus, dedicating more detailed analysis only to the most promising classification paths.

Stage 3: Aggregation and “Other” Handling Finally, the top-level Manager Agent aggregates all the confirmed predictions from both stages. The final output is the set of all parent narratives and child sub-narratives that were successfully proposed and then validated by their specialist agents. This workflow inherently enforces the true-path rule. If the top-level Manager proposes no narratives, or if none of its proposals are confirmed by the specialist agents, the document is automatically assigned the “Other” label.

Prompt engineering for zero-shot classification

The performance of the agentic framework is critically dependent on the quality and precision of the system prompts that guide each agent’s behavior. In a zero-shot setting, the prompt is the sole source of instruction, defining the agent’s role, task, and decision-making criteria.

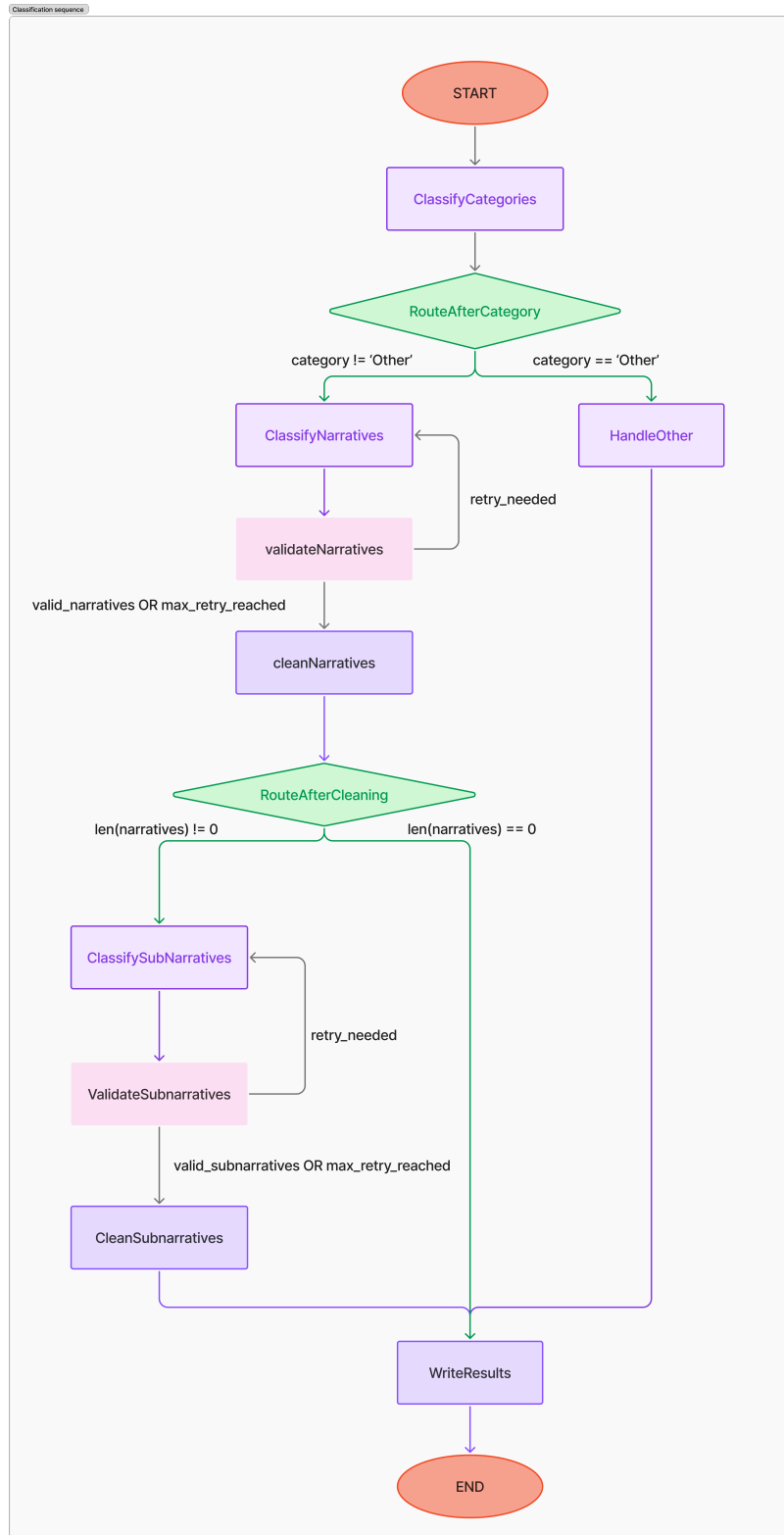


Figure 4.3: Actor–Critic pipeline: a stepwise, evidence-grounded workflow. The Actor generates structured claims grounded in verbatim evidence; the Critic validates these claims and may route feedback to the Actor for iterative refinement. Sub-narrative Actors and Critics repeat this process hierarchically.

Stage 4: The Narrative Critic (Claim Validation) The output of the Actor is immediately passed to the “Critic,” a separate LLM agent with a distinct, skeptical persona. The Critic’s sole purpose is to validate the Actor’s claims against a strict set of criteria: evidence accuracy (is the quote verbatim?), relevance (does the quote directly support the narrative?), and completeness. If the Critic finds the Actor’s analysis flawed, it generates structured feedback detailing the errors. The graph’s conditional logic then routes the state back to the Actor for a retry, passing the Critic’s feedback to a refinement prompt that instructs the Actor to correct its previous mistakes. This feedback loop can be repeated, creating a self-correction mechanism.

Stages 5 & 6: Sub-narrative Actor and Critic Once a set of narratives has been approved by the Critic, the process is repeated hierarchically for the sub-narrative level. For each validated parent narrative, a dedicated Sub-narrative Actor generates evidence-grounded claims for its children. To improve efficiency, these claims are generated in a batch operation. The claims are then passed to a Sub-narrative Critic for validation, which also has the ability to trigger a retry loop.

Stage 7: Finalization After passing through all stages, the final set of validated narratives and sub-narratives is written to the output file.

Configurable validation modes (experimental settings) The pipeline exposes narrative- and subnarrative-level critique validation as two independent, configurable switches in the YAML specification. In principle, both validation stages can be enabled sequentially (narratives validated before generating or validating subnarratives). However, to isolate marginal effects and avoid interaction confounds, our ablation in Section 6.4 evaluates the three mutually exclusive modes:

1. **No Validation:** Actor outputs at both hierarchy levels are accepted directly (Critic nodes disabled).
2. **Narrative-Only Validation:** A Critic validates and (optionally) triggers refinement loops for narrative claims; subnarrative claims are taken from the initial Sub-narrative Actor without critique.

Subnarrative classification. Most importantly for our primary F1 Samples metric, subnarrative classification significantly benefits from higher thresholds than the training baseline. Threshold 0.8 achieves the best F1 Samples performance (0.1750, representing a +13.4% improvement over the training threshold), while also excelling in F1 Micro (0.1574, +10.3% improvement). Threshold 0.7 provides the best F1 Macro (0.0821, +5.9% improvement). This pattern suggests that more conservative predictions are particularly beneficial for fine-grained classification when evaluated on a per-document basis, likely because they reduce false positives in the challenging task of distinguishing between subtle subnarrative categories.

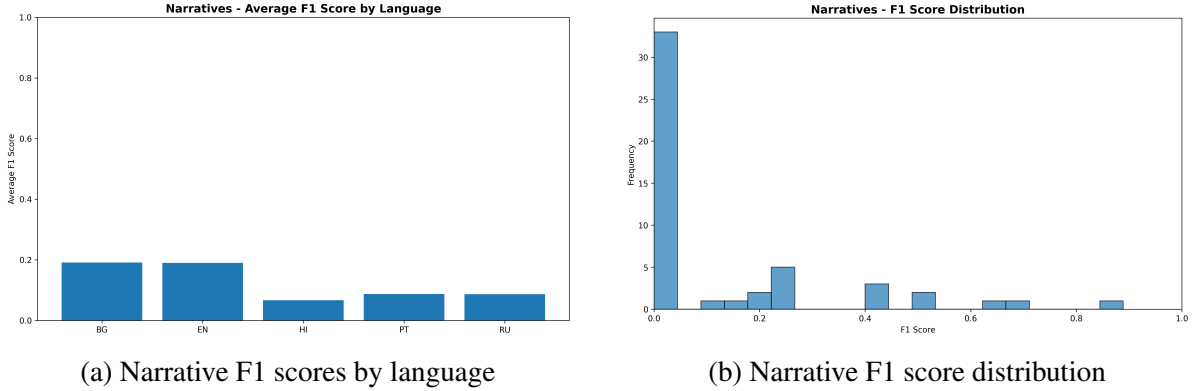


Figure 6.1: Narrative classification performance analysis at optimal threshold (0.65). (a) shows language-specific performance variations, with Portuguese achieving the highest F1 scores. (b) displays the distribution of F1 scores across all narrative categories, revealing the challenge of class imbalance.

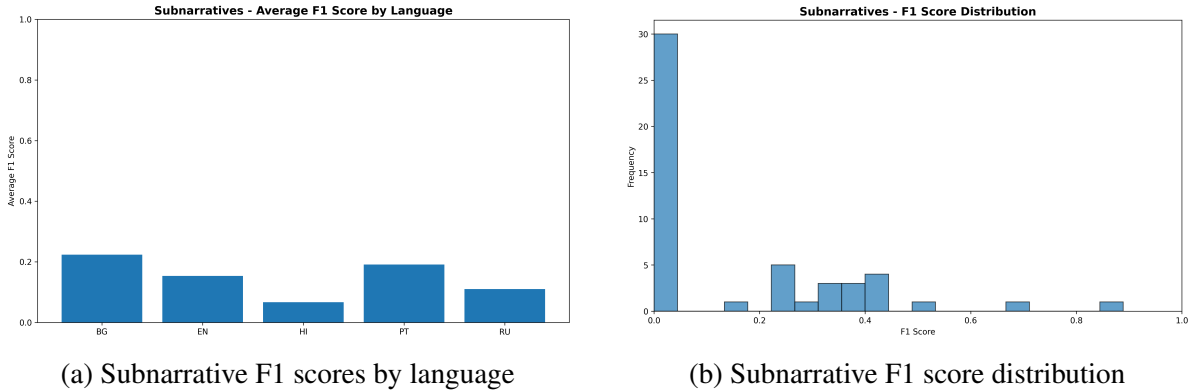


Figure 6.2: Subnarrative classification performance analysis at optimal threshold (0.7). (a) reveals significant language-specific challenges, particularly for Hindi. (b) shows the severely skewed F1 distribution, with most subnarratives achieving very low performance due to data scarcity.

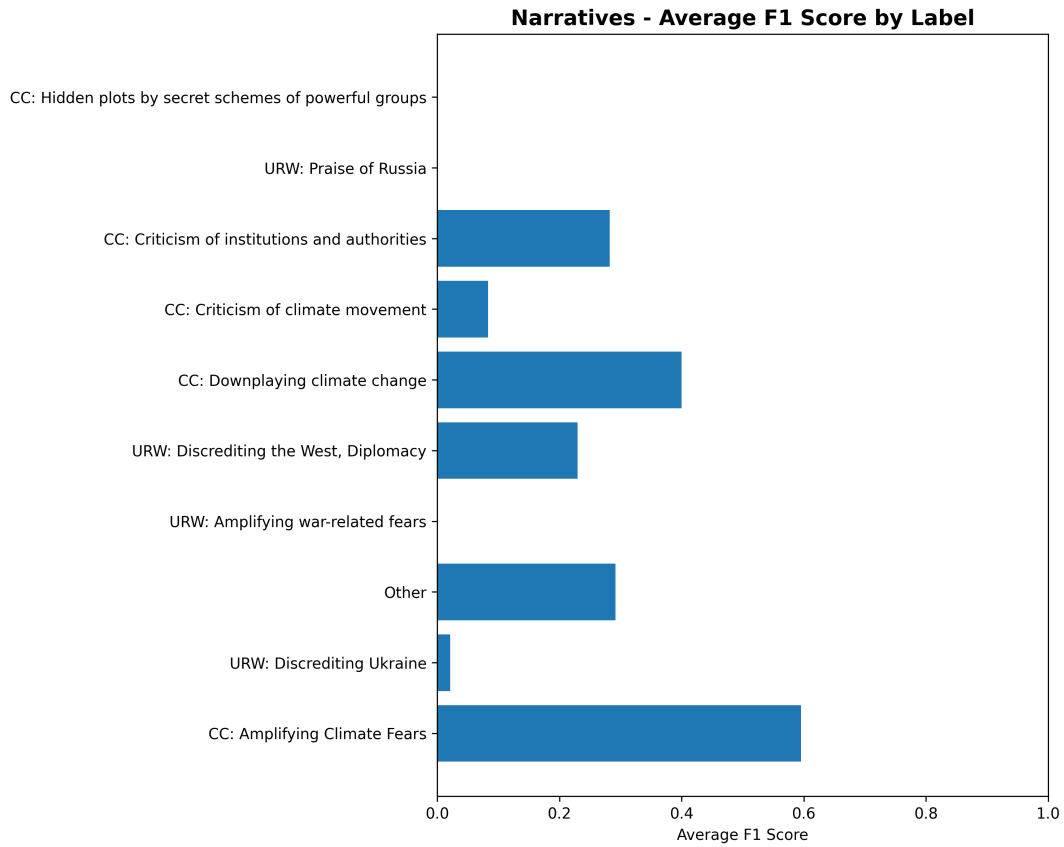


Figure 6.3: F1 scores for individual narrative categories at threshold 0.65. Climate change narratives (CC:) generally outperform Ukraine-Russia war narratives (URW:), with “CC: Questioning measurements and science” achieving the highest performance. The substantial variation across categories highlights the impact of class imbalance and linguistic complexity.

6.3.2 Multilingual Performance Analysis

To assess the framework’s cross-lingual capabilities, we conducted additional experiments on the Portuguese (PO) and Russian (RU) subsets. Our strategy involved translating the source documents into English before passing them to the agentic system. The results, presented in Table 6.5, show a significant performance degradation compared to the native English results.

Table 6.5: Multilingual performance comparison. The “translate-to-English” strategy results in a notable drop in performance.

Language	F1 Macro (Coarse)	F1 Samples (Fine)
English (Native)	0.513	0.406
Portuguese (Translated)	0.285	0.173
Russian (Translated)	0.247	0.137

The Sample F1 score for Portuguese (0.173) and Russian (0.137) is less than half of that achieved on the native English text. This suggests that the “translate-then-classify” approach, while straightforward, is a significant bottleneck. The translation process likely loses critical cultural and contextual nuances, rhetorical devices, and subtle linguistic cues that are essential for accurate narrative detection. This finding underscores the limitation of relying on a single-language system for a truly multilingual task.

6.3.3 Critical implications for transformer baseline methodology

Important Methodological Insight

These translation-induced performance degradations raise **significant concerns about our data augmentation methodology** used for the transformer baseline. The systematic performance drops (44–66% reduction in F1 Sample scores) suggest that machine translation fundamentally alters the linguistic and cultural markers essential for propaganda detection.

Data augmentation strategy reconsidered. Our transformer baseline approach relied heavily on translating texts between languages within the same dataset to increase training data volume across languages. Given the substantial information loss observed in the translate-then-classify paradigm, this data augmentation process was likely **fundamentally flawed** and may

have negatively impacted the transformer baseline’s multilingual performance by introducing translation artifacts and degraded linguistic representations during training.

Hypothesis for performance degradation. If similar degradation occurred during the data augmentation process for transformer training, the baseline model may have learned from *corrupted representations* of multilingual propaganda texts, potentially explaining some of the performance limitations observed in our transformer experiments.

Future work recommendation. This hypothesis warrants investigation in future work through controlled experiments comparing transformer performance on native versus translation-augmented training data to isolate the impact of translation-based data augmentation on multilingual propaganda classification accuracy.

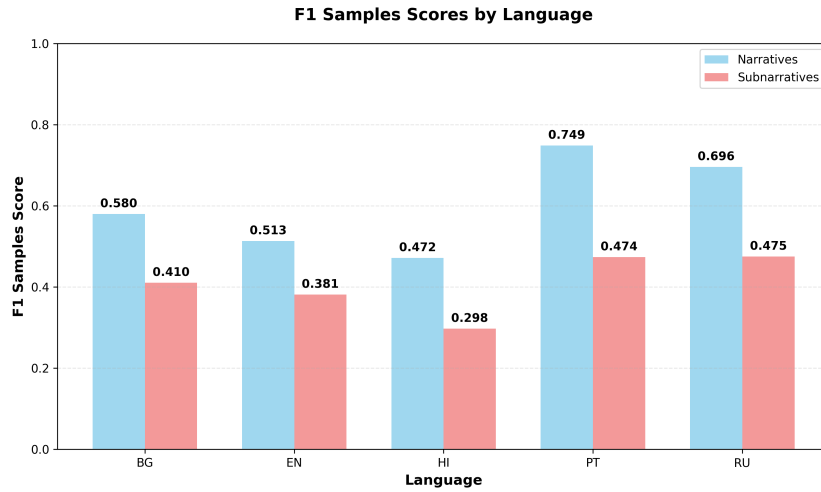
6.3.4 Detailed Error Analysis (on English DEV Set)

A detailed analysis of the model’s predictions on the English development set reveals two primary failure modes, which are characteristic of long-tailed, semantically complex classification tasks.

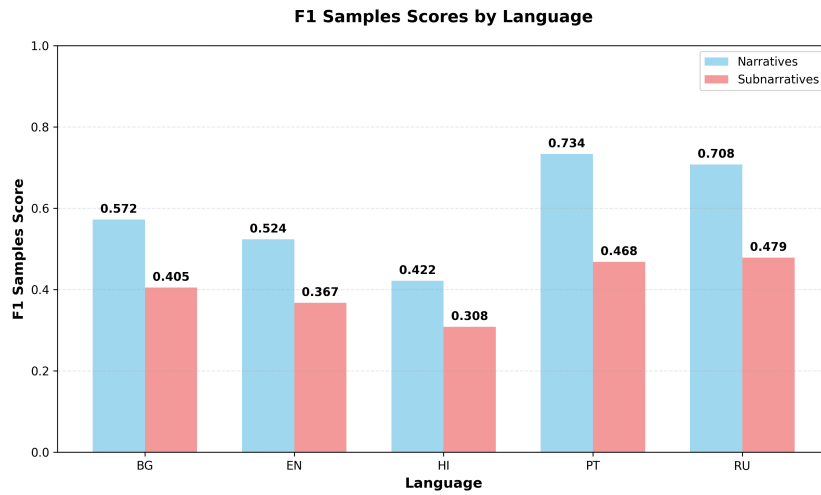
Systematic Failures on Rare Narratives

Despite the strong Macro F1 score, the model completely failed to identify any positive instances for several of the rarest narratives. On the development set, 9 out of 22 narratives (41%) had zero true positives. These included both climate-related narratives (e.g., “Amplifying Climate Fears”) and war-related narratives (e.g., “Russia is the Victim”). This indicates that while the zero-shot approach is not dependent on training examples, the LLM’s internal knowledge or the provided prompt definitions were still insufficient to detect these highly infrequent or subtly expressed narratives.

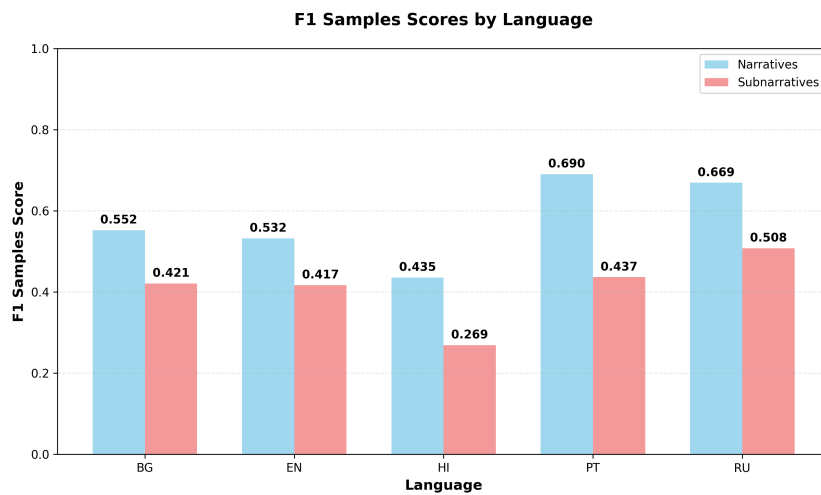
6 Experiments and results



(a) Without validation



(b) Narrative validation



(c) Subnarrative validation

Figure 6.4: Three-way comparative F1 Samples performance across languages and validation strategies. Each validation approach demonstrates distinct language-dependent effectiveness patterns, with Russian showing highest combined performance under subnarrative validation, Portuguese excelling with narrative validation, and Bulgarian achieving consistent improvements across both validation levels.

Sample by -0.0022, while subnarrative validation shows larger decreases of -0.0261 F1 Coarse with minimal improvement (+0.0025 F1 Sample). This counter-intuitive result suggests that the base Gemini 2.5 Flash model’s initial predictions are often more accurate than the validation-corrected versions, indicating potential over-correction or validation noise issues.

Language-specific validation sensitivity reveals mixed patterns. Russian is the only language that consistently benefits from narrative validation (+0.0122 F1 Coarse, +0.0032 F1 Sample), while all other languages show performance degradation. Portuguese experiences the most significant narrative-level decline with both validation approaches (-0.0154 narrative validation, -0.0585 subnarrative validation), despite having the highest baseline performance (0.7489 F1 Coarse). English demonstrates the most complex validation trade-offs: narrative validation provides modest narrative gains (+0.0109) but hurts subnarrative performance (-0.0140), while subnarrative validation improves both levels (+0.0187, +0.0351).

Hierarchical validation effects show unpredictable cross-level interactions. The validation mechanisms produce complex hierarchical interdependencies that often work against intended improvements. Hindi exemplifies this: narrative validation significantly damages narrative performance (-0.0498) while slightly improving subnarrative classification (+0.0106), suggesting that validation-induced narrative changes may inadvertently benefit fine-grained accuracy through error compensation rather than genuine improvement. This indicates that hierarchical validation effects are not predictably additive and may introduce systematic biases.

Computational overhead without systematic benefit. Both validation strategies introduce significant computational overhead through multi-stage processing while delivering marginal or negative performance gains across most languages and metrics. The largest individual improvements are modest (English subnarrative validation: +0.0351), while the largest degradations are substantial (Portuguese subnarrative validation: -0.0585), creating an unfavorable cost-benefit profile for validation mechanisms in this multilingual context.

6.4.5 Hierarchical performance characteristics

The actor-critique pipeline demonstrates a consistent pattern of performance degradation from narrative to subnarrative classification across all languages:

- **Average performance gap:** 19.4% reduction from F1 Coarse (narratives, 0.6019) to F1 Sample (subnarratives, 0.4077).
- **Smallest gap:** English (13.2% reduction).
- **Largest gap:** Portuguese (27.4% reduction).

This hierarchical performance pattern is significantly smaller than observed in the transformer baseline (60% reduction), suggesting that the actor-critique validation process effectively mitigates some error propagation issues inherent in hierarchical classification.

6.4.6 Language-specific narrative performance analysis

Figure 6.5 illustrates the top-performing narrative categories for each language, revealing distinct cross-linguistic patterns.

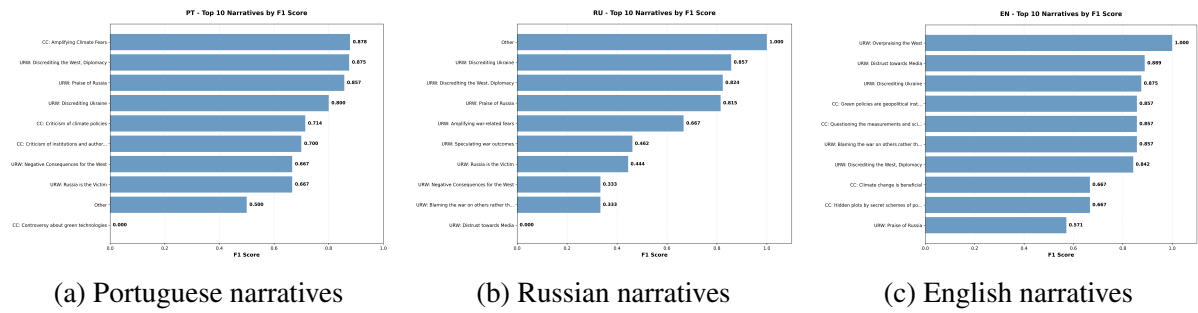


Figure 6.5: Top-performing narrative categories by language, showing the actor-critique pipeline’s effectiveness varies significantly across narrative types and languages. Portuguese and Russian show strong performance on climate and war narratives respectively, while English demonstrates more balanced performance across categories.

Portuguese narrative excellence

Portuguese demonstrates exceptional performance across climate change narratives:

- **CC: Amplifying Climate Fears:** F1 = 0.878 (perfect precision: 1.000)
- **URW: Discrediting the West, Diplomacy:** F1 = 0.875
- **URW: Praise of Russia:** F1 = 0.857 (perfect precision: 1.000)

The consistently high precision scores (many achieving 1.000) indicate that the Portuguese language processing benefits significantly from the actor-critique validation mechanism, effectively eliminating false positives.

Russian war narrative specialization

Russian text shows particular strength in war-related narratives:

- **Other:** Perfect performance (F1 = 1.000)
- **URW: Discrediting Ukraine:** F1 = 0.857 with high recall (1.000)
- **URW: Discrediting the West, Diplomacy:** F1 = 0.824
- **URW: Praise of Russia:** F1 = 0.815 with high precision (0.917)

This pattern suggests that the pipeline effectively captures Russian-specific linguistic markers for war-related propaganda narratives.

English balanced performance

English demonstrates more balanced performance across both climate and war narratives:

- Multiple perfect scores on specific categories (F1 = 1.000)
- Strong performance on technical climate narratives (CC: Questioning measurements and science: F1 = 0.857)
- Effective war narrative detection (URW: Discrediting Ukraine: F1 = 0.875)

6.4.7 Subnarrative classification analysis

Figure 6.6 presents the fine-grained classification performance, highlighting the challenges of semantic disambiguation at the subnarrative level.

6 Experiments and results

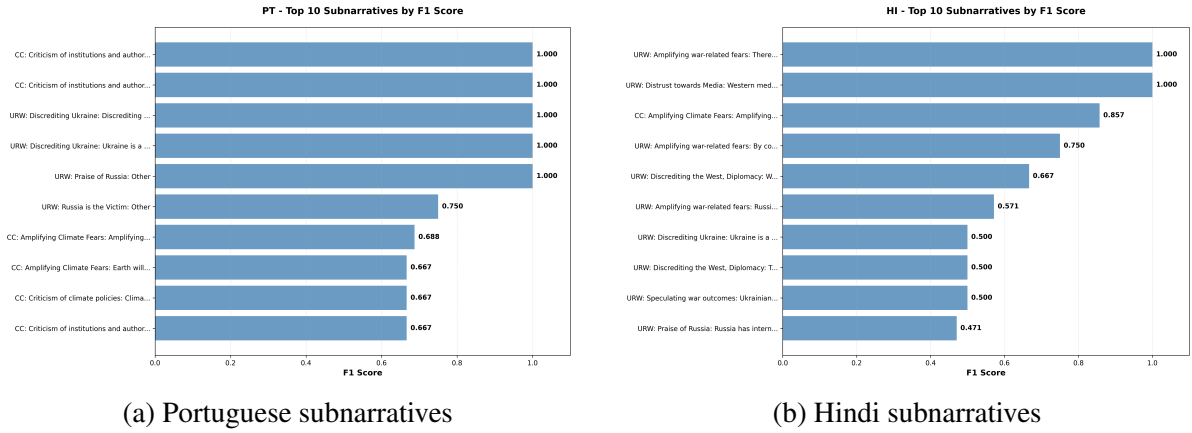


Figure 6.6: Subnarrative performance comparison between the best (Portuguese) and most challenging (Hindi) languages. Portuguese maintains strong performance even at fine-grained levels, while Hindi shows the characteristic challenges of low-resource language processing in nuanced classification tasks.

Portuguese maintains fine-grained excellence

Portuguese achieves remarkable performance even at the subnarrative level:

- Multiple perfect classifications ($F1 = 1.000$) across diverse categories
- **CC: Amplifying Climate Fears: Amplifying existing fears:** $F1 = 0.688$ with strong recall (0.786)
- Consistent high precision across categories, indicating effective false positive control

Hindi subnarrative challenges

Hindi reveals systematic challenges at the subnarrative level:

- Best performance: **URW: Amplifying war-related fears: Real possibility of nuclear war** ($F1 = 1.000$)
- **CC: Amplifying Climate Fears: Amplifying existing fears:** $F1 = 0.857$
- Significant performance degradation on most categories, with many achieving $F1 < 0.500$

6.4.8 Comparative analysis with baseline architectures

All three actor-critique configurations demonstrate substantial improvements over the transformer baseline, with optimal validation strategy selection depending on specific language and performance requirements:

Hierarchical consistency marginally improved despite performance degradation.

While validation strategies generally reduce absolute performance, they do provide slight improvements in hierarchical consistency: no validation shows a 19.4% performance gap between levels (0.6019 vs 0.4077), narrative validation reduces this to 18.6% (0.5919 vs 0.4055), and subnarrative validation achieves 16.8% (0.5758 vs 0.4102). However, this marginal consistency improvement comes at the cost of overall performance reduction, questioning whether hierarchical alignment justifies the performance trade-off.

No-validation configuration achieves optimal overall performance. Contrary to expectations, the simplest configuration without validation delivers the highest average performance across both hierarchical levels (F1 Coarse: 0.6019, F1 Sample: 0.4077). Portuguese exemplifies this pattern most clearly, achieving its peak narrative performance (0.7489) without validation, with both validation approaches causing substantial degradation. This suggests that Gemini 2.5 Flash’s base predictions are often more reliable than validation-corrected outputs.

Validation mechanisms introduce systematic over-correction bias. The consistent performance degradation across most language-validation combinations indicates that the critique agents may be systematically over-correcting the base model’s predictions. This is particularly evident in Portuguese, where both validation strategies cause significant performance drops (-0.0154 to -0.0585), and in Hindi, where narrative validation severely damages narrative classification (-0.0498) despite the language having the lowest baseline performance and theoretically the most room for improvement.

6.4.9 Key insights and architectural implications

Simplicity outperforms complexity in multilingual hierarchical classification.

Our comprehensive ablation study provides strong evidence that the no-validation configuration

Overall Performance Comparison

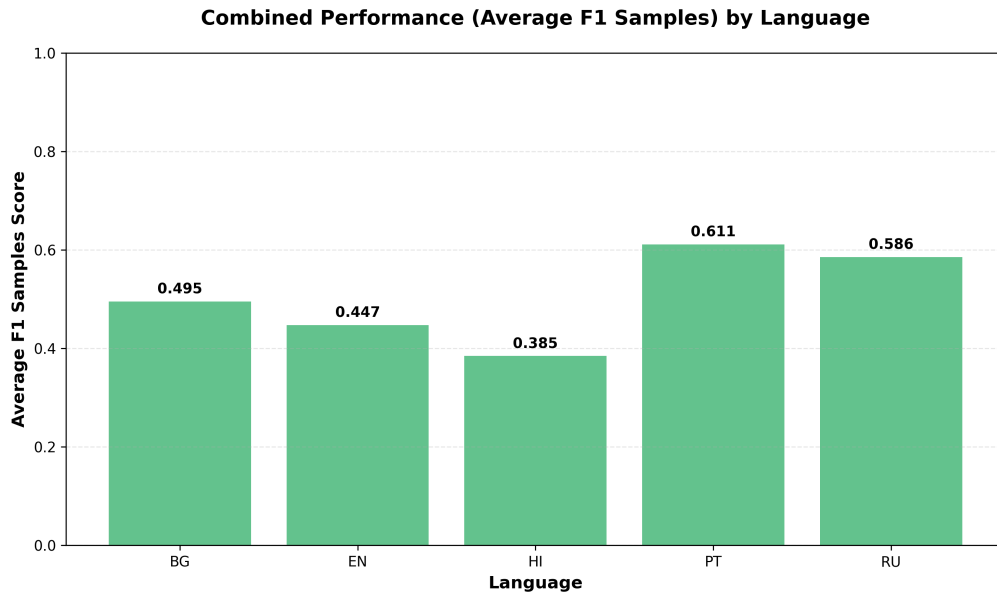


Figure A.1: Combined performance across all languages for the no-validation configuration, showing F1 Coarse (narratives) and F1 Sample (subnarratives) metrics.

A.1.2 Bulgarian (BG) Performance

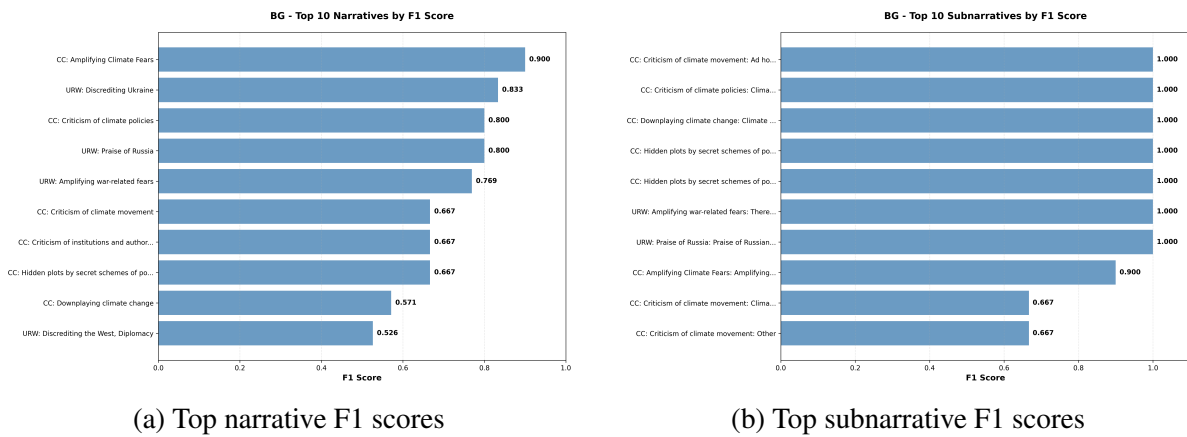


Figure A.2: Bulgarian performance breakdown for no-validation configuration.

A.1.3 English (EN) Performance

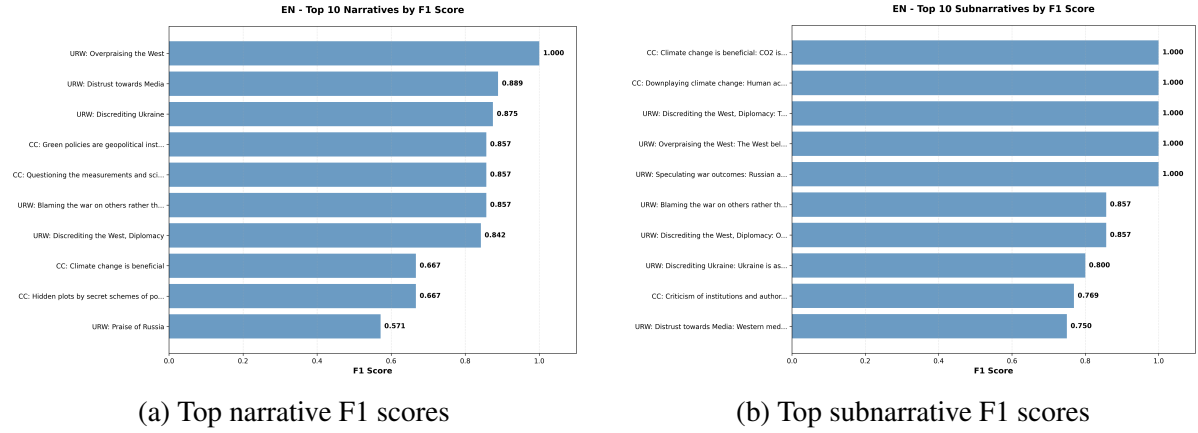


Figure A.3: English performance breakdown for no-validation configuration.

A.1.4 Hindi (HI) Performance

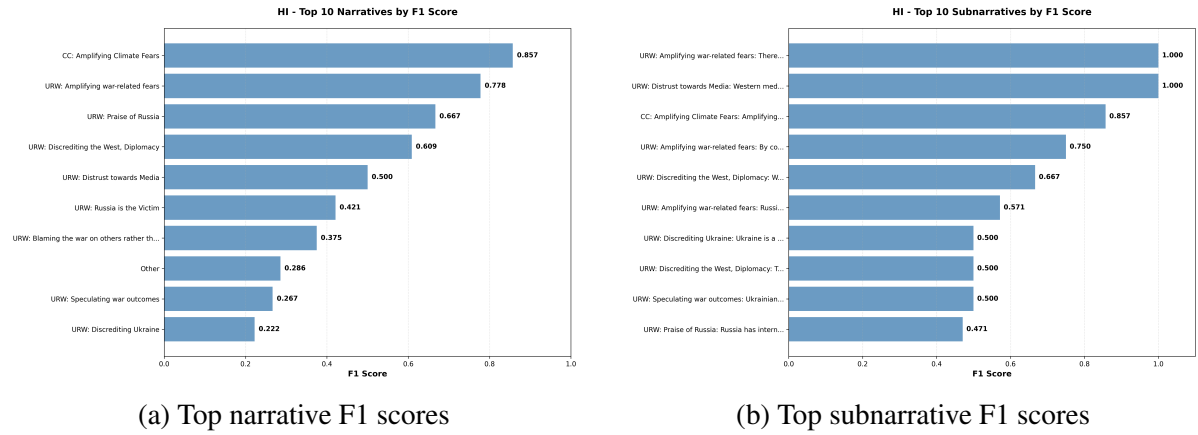


Figure A.4: Hindi performance breakdown for no-validation configuration.

A.1.5 Portuguese (PT) Performance

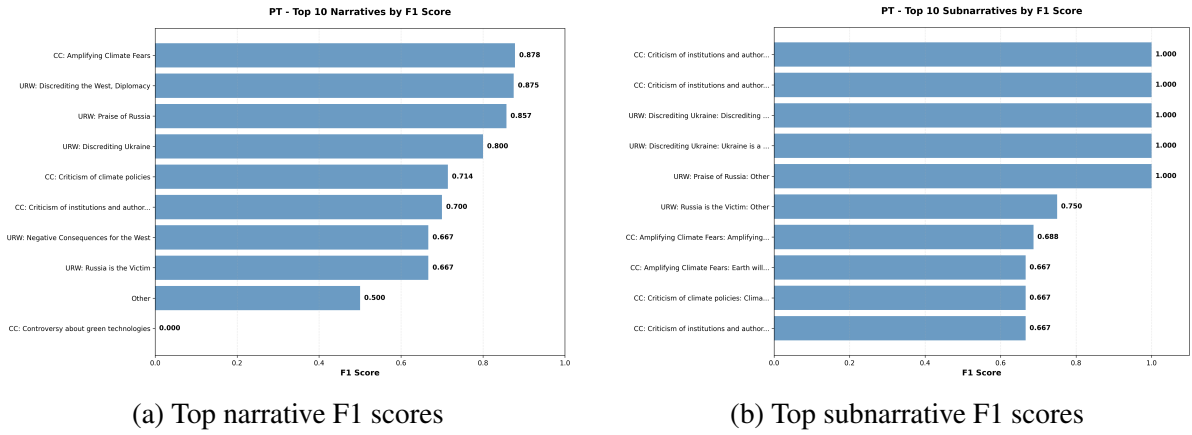


Figure A.5: Portuguese performance breakdown for no-validation configuration.

A.1.6 Russian (RU) Performance

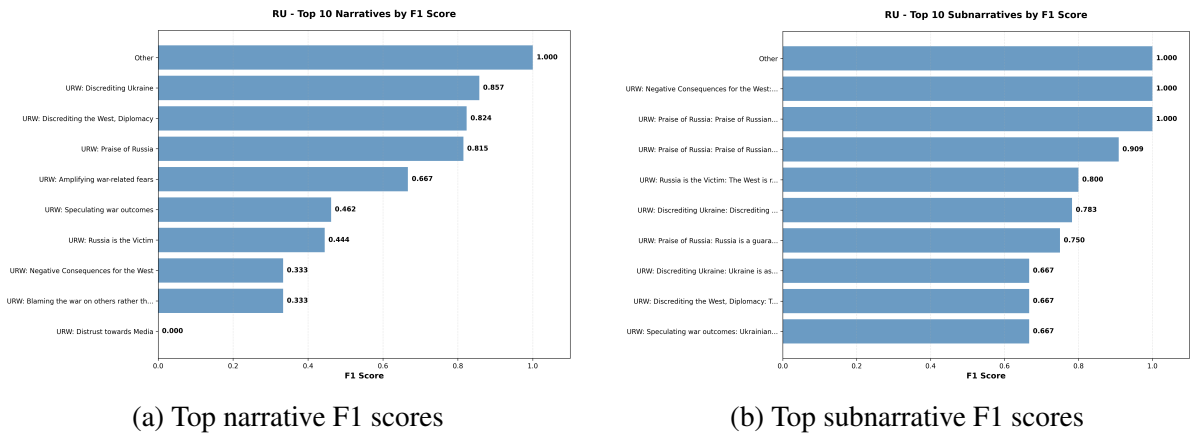


Figure A.6: Russian performance breakdown for no-validation configuration.

A.1.7 A.2 Narrative Validation Configuration

This section presents representative results from the Actor-Critique pipeline with narrative-level validation enabled. Complete per-language breakdowns follow the same patterns as the no-validation configuration.

A.1.8 Overall Performance Comparison

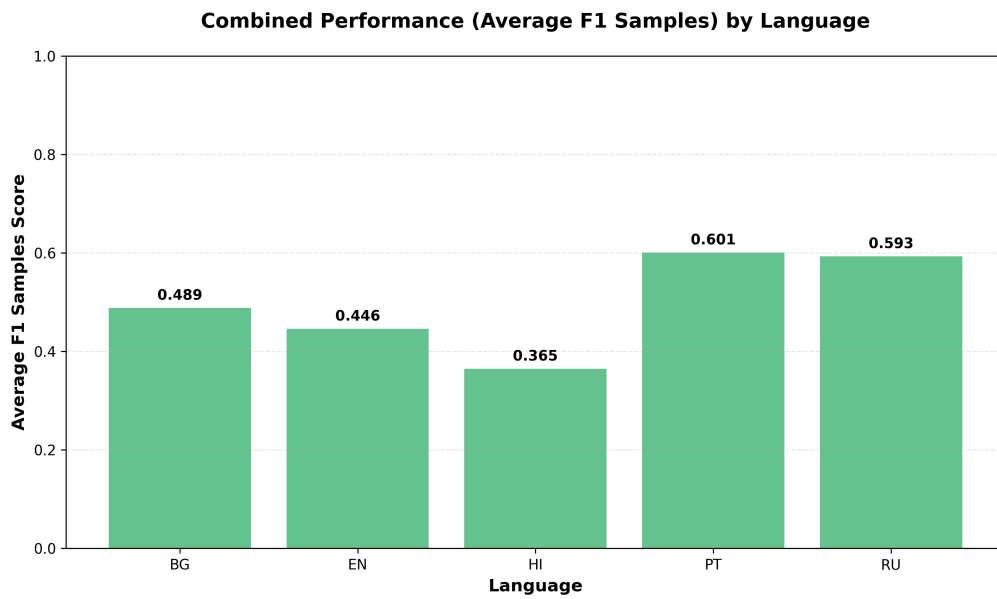


Figure A.7: Combined performance across all languages for the narrative-validation configuration.

A.1.9 Representative Example: English (EN) Performance

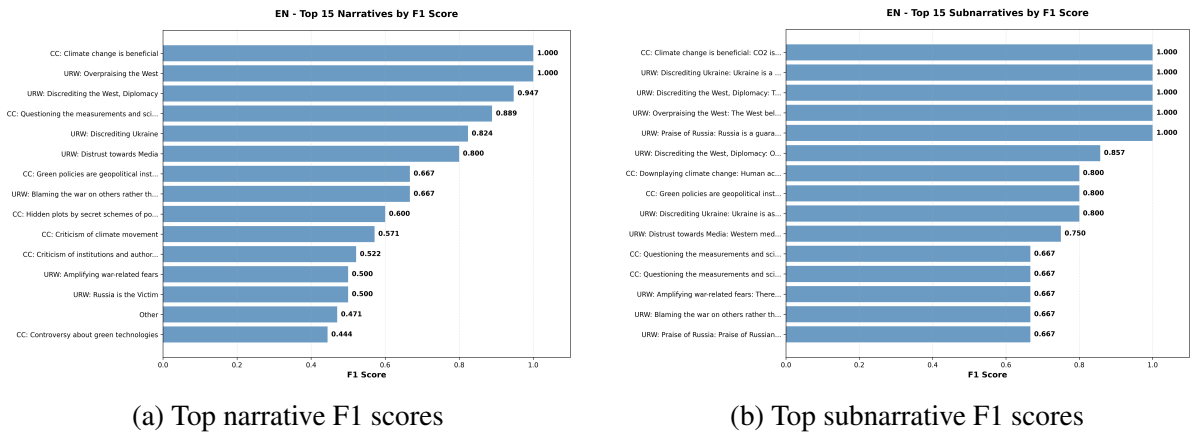


Figure A.8: English performance breakdown for narrative-validation configuration, representative of per-language patterns.

A.1.10 A.3 Subnarrative Validation Configuration

This section presents representative results from the Actor-Critique pipeline with subnarrative-level validation enabled. Complete per-language breakdowns follow the same patterns as the no-validation configuration.

A.1.11 Overall Performance Comparison

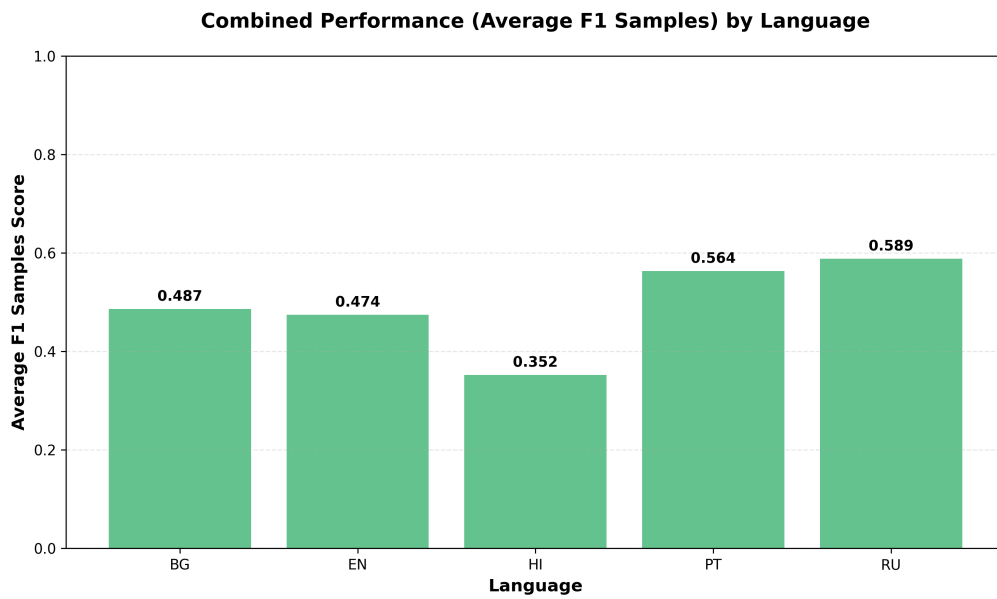


Figure A.9: Combined performance across all languages for the subnarrative-validation configuration.

A.1.12 Representative Example: English (EN) Performance

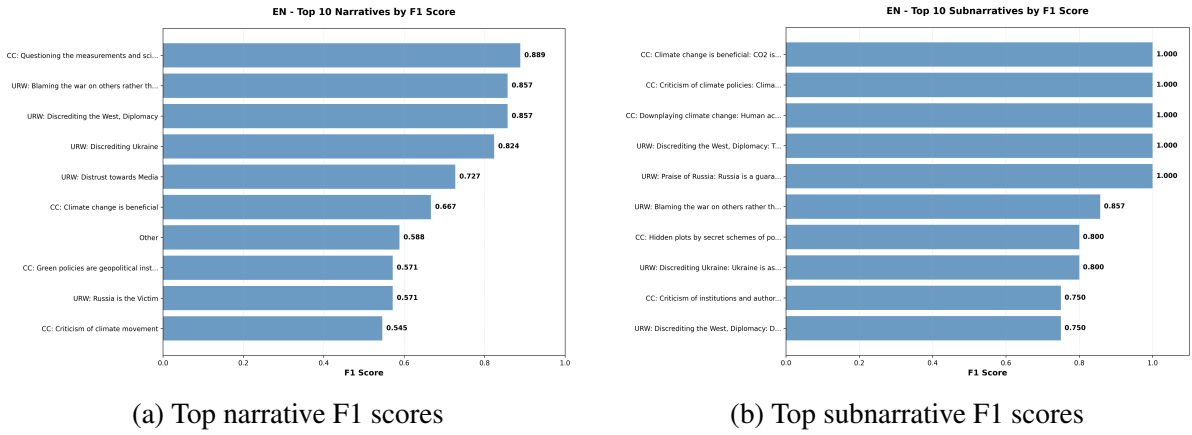


Figure A.10: English performance breakdown for subnarrative-validation configuration, representative of per-language patterns.

A.2 Appendix B: Prompt Templates

This section provides the complete prompt templates used in both Architecture 1 (AutoGen Multi-Agent System) and Architecture 2 (Actor-Critique Pipeline). The prompts demonstrate the different approaches: Architecture 1 uses simple binary classification prompts, while Architecture 2 employs more sophisticated chain-of-thought reasoning with structured JSON output.

A.2.1 B.1 Architecture 1: Binary Classification Prompts

Architecture 1 employs a multi-agent system where each agent performs binary classification (0 or 1) to detect the presence of a specific narrative or subnarrative.

Narrative Detection Prompt

Architecture 1: Narrative Binary Classification

You are a classification model trained to do binary classification by detecting whether a given text is related to a specific narrative or not.

You have been trained to recognize the narrative:

NARRATIVE.

defined as:

NARRATIVE_DEFINITION.

Here are some examples of statements related to this narrative:

NARRATIVE_EXAMPLES.

If the text is related to the narrative, you MUST respond with '1' only.

Otherwise, you MUST with '0' only.

You are ONLY allowed to answer with '1' or '0' and NOTHING else.

Only answer with 1 if there are EXPLICIT and CLEAR mentions of the narrative in the text. Some text will be ambiguous so if you are slightly